

Exploring the knowledge and application of biosecurity practices related to invasive alien species in field research in Croatia

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ABSTRACT

Accidental introductions of invasive alien species (IAS) are increasingly prevalent and costly. Anthropogenic activities, including fieldwork activities of researchers, pose a risk for the unintentional spread of IAS. However, specific biosecurity policies and measures to prevent accidental introductions and the secondary spread of IAS are lacking in many European countries, primarily relying on researchers' awareness of the issue. This study aimed to evaluate the biosecurity practices of researchers in Croatia during their fieldwork activities through an anonymous online survey distributed to over 63 institutions. The survey was completed by 172 respondents from 45 institutions and aimed to assess their knowledge of IAS and awareness of biosecurity risks they can pose. The results showed that even though more than half of the participants worked in the field of biology and had at least some knowledge of IAS, the majority used only the most basic and least efficient cleaning approaches for fieldwork equipment, clothes, and footwear, which are not sufficient for preventing unintentional IAS introductions. Moreover, vehicle cleaning was the most neglected biosecurity measure in our study, with up to 60 % of respondents never performing it and only 12 % always performing at least some type of vehicle cleaning. Respondents with the best knowledge of IAS posed a significant risk during their fieldwork activities but at the same time demonstrated the best understanding of biosecurity measures and had the best biosecurity performance. Furthermore, students were identified as a high-risk group since they visited the highest number of locations during fieldwork and used the equipment most often but cleaned it the least frequently. Based on obtained results, we discuss potential interventions that could help manage accidental IAS introductions at the national and EU levels.

1. Introduction

Global biodiversity, ecosystem services, and human well-being are at risk due to biological invasions and the spread of invasive alien species (IAS) (Turbelin et al., 2023). The costs of incurred damage and the required IAS management actions are of the same magnitude order as natural hazards, yet their threat is much less perceived by the general public and responsible institutions (Turbelin et al., 2023).

Overcoming natural barriers as a result of human activity has allowed IAS to spread and proliferate (Pyšek et al., 2020). According to research on global invasion and introduction patterns of IAS, 39 % of 1517 IAS documented in 243 countries and overseas territories were introduced intentionally, as opposed to 26 % of IAS that were accidentally introduced (Turbelin et al., 2017). However, the proportion of

accidentally introduced IAS may be much higher: on a country-level it was shown to range between 25 % and up to 100 % (Turbelin et al., 2017). Furthermore, in 63 % of the world's nations, accidentally introduced species outnumber the intentionally introduced ones. Additionally, the economic costs incurred by accidentally introduced IAS (i.e., damage and management costs) have also been assessed as four times higher than the cost needed to cover the damage caused by intentionally introduced species (Turbelin et al., 2017). This highlights the need to effectively detect and manage accidental introductions of species.

Existing policies (i.e., Convention on Biological Diversity (CBD, 2014), EU IAS Regulation 1143/2014) and IAS experts agree that preventing both intentional and unintentional introduction of IAS is the most efficient and cost-effective approach to managing biological invasions. However, pathways of accidental introduction are difficult to

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manage. Out of the six identified primary introduction pathways for IAS, five are considered unintentional: escape after intentional introduction, introduction as a contaminant with a particular commodity, introduction as a stowaway on a transport vector, introduction via corridors, and unaided introduction through natural dispersal (Hulme et al., 2008). When it comes to accidentally introduced organisms, identifying introduction pathways is often difficult since it is frequently a consequence of ubiquitous and diverse human activities, mainly the trading of goods, transportation, and recreation (Keller et al., 2011; Hulme, 2015; Saul et al., 2017; Hanrahan and Melly, 2019; Smith et al., 2020; Melly and Hanrahan, 2020, 2023).

Accidental IAS introductions and their pathway management requires undertaking biosecurity actions and measures (Shannon et al., 2020a). However, the implementation of biosecurity measures requires a systematic approach integrated within national and international policies and regulatory frameworks. Additionally, it necessitates changes in human behaviour, which can only be achieved through a combination of diverse interventions ranging from education and awareness raising to various policy measures (Shannon et al., 2020a; Shannon et al., 2020b). Unfortunately, biosecurity policies regarding IAS are still limited, or even completely lacking in many European countries. Member states significantly differ in their regulation policies regarding the control of accidental introductions. A recent review by Baquero et al. (2021a) identified sectoral fragmentation of regulatory policies aimed at protecting biodiversity at Natura 2000 sites as one of the major obstacles to managing IAS. Moreover, the IAS Regulation (EU Regulation 1143/2014) covers only a small proportion of all IAS present in the EU (Baquero et al., 2021b), while biosecurity is mostly addressed through sectoral voluntary EU codes of conduct related to invasive species (EASIN, 2023). In Croatia, action plans for control of unintentional introduction pathways of IAS by transport and natural dispersal have been established in 2022 (MINGOR, 2022). Within these plans, major stakeholders have been identified, and educational activities and actions involving biosecurity issues are planned to take place over the next 10 years.

Here, we aim to determine the awareness of biosecurity risks posed by fieldwork activities of researchers in Croatia, a group of individuals who should be relatively familiar with the issue of IAS transmission. They also frequently and regularly conduct both biodiversity and habitat monitoring, which involves biosecurity risks and may result in accidental transmission of IAS. While IAS educational campaigns have been launched recently within the frame of the project “Development of management and control systems for IAS”, no biosecurity campaigns have been conducted in Croatia so far. Generally, awareness of IAS issues has been identified as poor in the whole region (Piria et al., 2021). Therefore, we aimed to examine the extent to which researchers in Croatia implement biosecurity measures during fieldwork.

2. Methods

2.1. Survey design, adaptation, and distribution

The research was conducted in April of 2021 through an anonymous online survey in Croatia using the Google Form platform. For the purpose of this study, we adapted the survey developed by Shannon et al. (2019).

The modifications made to the questionnaire of Shannon et al. (2019) included the removal of questions that do not apply to our study area, the shortening of certain questions, and the addition of IAS photographs characteristic of Croatia for identification in the last section of the questionnaire (Appendix A). A list of all question groups is presented in Table 1, and the complete survey is available in Appendix A. Following Shannon et al. (2019), the survey consisted of an introductory section (gender, age, institution and role in it, disciplinary field) and ten groups of questions that examined fieldwork activities, sampling methods and habits, equipment usage, as well as frequency and

Table 1

Description of question groups from the survey.

Group No.	Question groups	Question description
1	Basic information	Information on age, disciplinary field, institution and position.
2	Locations for the implementation of field activities	Locations of field activities (Croatia and/or other countries), frequency of sampling in protected areas, types of habitats in which samples are collected.
3	Field activity	Number of locations visited per field trip, role during the field trip, type of activities and sampling procedures, transfer of the samples between different locations.
4	Field equipment	Use of the equipment during fieldwork, contact between equipment and vegetation, frequency of equipment cleaning, methods used for equipment cleaning.
5	Transport	Means of transportation to field location, frequency of off-road driving, frequency of tires/wheels cleaning.
6	Field footwear and clothing	Frequency and methods of footwear and clothing cleaning.
7	Invasive alien species	Connection of field activities with IAS, self-perceived risk of spreading IAS while doing field activities.
8	Biosecurity knowledge	Familiarity with guidelines or campaigns on biosecurity, previous participation in biosecurity education.
9	Biosecurity measures	Applying biosecurity measures during field activities, planning activities to reduce the risk of spreading IAS.
10	IAS perceived knowledge	Identification of invasive alien species in Croatia through self-assessment.
11	IAS identification success	Identification of invasive alien species present in Croatia from photographs, either taken by authors or downloaded from Wiki Commons and Pixabay (photo credit added to Appendix A file, which was not visible to respondents in the survey).

approaches to cleaning field equipment, clothing, footwear, and means of transport. We also examined the frequency of participants' encounters with IAS in the field and the familiarity of survey participants with IAS topics, biosecurity guidelines, and campaigns, as well as their application of biosecurity protocols in the field. The last group of questions examined the respondents' knowledge of IAS, both through self-assessment and their success in identifying IAS through provided photographs.

The survey was performed online by sending a link with the questionnaire via e-mail to identified target groups (various institutions that were identified as those that perform fieldwork during their research) and was also distributed via mailing lists of conservation/nature protection experts and biology students in Croatia. These institutions included faculties of science, biotechnology, forestry, agriculture and all related disciplines, as well as public and municipal administrations in the field of environment and nature protection (including national and nature parks, museums, and aquariums), expert and student associations, social networks, and nature/environmental protection companies and non-governmental organizations in Croatia (see Table B1 in Appendix B).

2.2. Survey processing and data analysis

Since the survey consisted of different types of questions, they were processed according to their type. In multiple-choice questions, the score was equal to the number of choices that respondents selected (one selection is equivalent to one point), while yes/no questions were assigned a binomial score (0 or 1). Answers related to the frequency of

events were assigned scores from 1 (never) to 5 (always). Questions requiring a numeric answer were assigned a score based on the number of elements enumerated by the respondents. For each question group set out in Table 1, scores of all questions were summed, and the sum of scores was used as a measure of potential risk. The higher sum of scores in question groups related to the frequency and type of field activities, as well as those related to the use of equipment (question groups 2–4; Table 1; Appendix A) indicated a higher risk for the unintentional spread of invaders. This was due to factors such as higher number of locations visited during one fieldwork, visiting multiple regions during research, and frequent use of equipment that comes in contact with various types of environments (i.e., water, soil, vegetation). A higher sum of scores for question groups related to the cleaning of equipment, transportation (means of transport), clothing and footwear cleaning (question groups 5 and 6; Table 1; Appendix A), as well as those related to biosecurity measures, perceived knowledge, and identification performance (question groups 7–11; Table 1; Appendix A) represented a lower risk. In this context, higher scores corresponded to a higher frequency and number of methods used for cleaning field equipment, transportation, clothes, and footwear, as well as a higher knowledge level about IAS and higher IAS identification success.

All acquired scores for each question group were presented as proportions (%) of the total number of respondents (Figs B1–B27 in Appendix B). Differences in analysed question groups between age groups, and between different levels of exposure to IAS topics, as well as differences between respondents in biosecurity knowledge and IAS identification success, were analysed using nonparametric statistics (Kruskal–Wallis ANOVA with Dunn’s post-hoc test) (Peck et al., 2015; Donges, 2018). Correlations between the equipment use, cleaning of equipment, and fieldwork clothing, and the focus of fieldwork on IAS (question “How often do issues related to invasive alien species arise in relation to your field activity?”), as well as correlations between the biosecurity knowledge, perceived IAS knowledge, IAS identification success, and the focus of fieldwork on IAS, were analysed using Spearman rank-order correlation. Statistical data processing was performed in Microsoft Excel for Microsoft 365 MSO (16.0.14026.20294) and Statistica 13.5.0.17 (TIBCO Software Inc., USA). The significance level threshold was set at 0.05.

3. Results

An online anonymous survey was completed by 172 respondents from 45 institutions. The age structure, institution affiliation types, and expertise levels are presented in Appendix B. The sample was considered representative for experts performing fieldwork in Croatia, since it encompassed 7 out of 9 Universities in Croatia, major research institutes, public bodies for nature protection from 13 out of 21 counties in Croatia, experts from responsible ministries and sectoral agencies, private companies working on environmental protection, non-governmental institutions, museums and aquariums (Table B1 in Appendix B). Nevertheless, since over a half of the respondents were working in the field of biology (Fig. B5 in Appendix B), data were biased towards the professions which are expected to also have the highest awareness of IAS and biosecurity. Most respondents (56 %) were students and early career experts within the age groups of 18–25 & 26–35 (Fig. B2 in Appendix B). The largest segment of respondents was affiliated with the academic sector (universities and research institutes; Fig. B3 in Appendix B). Within this sector, over 72 % of respondents was in the fields of biology, ecology, and environmental science and protection (Fig. B5 in Appendix B).

3.1. Biosecurity measures, knowledge and IAS identification

More than a half of the respondents (54 %) believe that their field activities do not pose a risk in terms of the spread of IAS, while 17 % believe that their field activities do pose a risk (Fig. B21 in Appendix B).

The Kruskal–Wallis ANOVA with Dunn’s post-hoc test analysis of the question “Do you consider your field activities as a threat in terms of spreading IAS?” shows statistically significant difference in values of respondents who answered “Yes” to the question are those engaged in more field research; consequently, they present a greater risk in terms of spreading IAS ($H_{(2,172)} = 10.949$, $p = 0.004$; Fig. 1A). However, they also exhibit a better knowledge of biosecurity risks and practices ($H_{(2,172)} = 13.713$, $p = 0.001$; Fig. 1B), a higher perceived knowledge of IAS ($H_{(2,172)} = 10.919$, $p = 0.004$; Fig. 1C), and greater success in identifying IAS ($H_{(2,172)} = 10.975$, $p = 0.004$; Fig. 1D) compared to the other two groups (respondents who answered “No” or “Not sure”).

Only a small fraction of respondents (29 %) is aware of biosecurity guidelines or campaigns, or have participated in any form of biosecurity education (24 %; Figs B22, B24 in Appendix B). The analysis of respondents’ knowledge by identifying seven IAS revealed that only 15 % correctly recognized all species as IAS. The Asian tiger mosquito (*Aedes albopictus*) was the most accurately recognized IAS (90 %), while nutria (*Myocastor coypus*) was the least accurately identified IAS, with 49 % identifying it as invasive and 34 % mistakenly classifying it as non-invasive species (Fig. B27 in Appendix B).

No significant differences were established between age groups and biosecurity knowledge, perceived knowledge of IAS, and identification success of IAS. However, a significant positive correlation was observed between the focus of fieldwork on IAS and biosecurity knowledge ($r_s = 0.195$, $p < 0.05$), biosecurity practices ($r_s = 0.463$, $p < 0.05$), perceived knowledge of IAS ($r_s = 0.434$, $p < 0.05$), and identification success ($r_s = 0.260$, $p < 0.05$). Biosecurity practices were also positively correlated with perceived IAS knowledge ($r_s = 0.403$, $p < 0.05$) as well as biosecurity knowledge ($r_s = 0.198$, $p < 0.05$).

3.2. Sampling locations, field activities and equipment use

The majority of respondents (90 %) carry out their fieldwork in protected areas at least occasionally. Additionally, 65 % of respondents visit only one site per field trip (Figs B6, B9 in Appendix B). Most respondents collect samples (usually more than one type) during fieldwork, primarily live specimens and water samples (Figs B11, B12 in Appendix B), which are then transferred between sampling sites during the fieldwork (Fig. B13 in Appendix B).

Most respondents (94 %) use the equipment during their fieldwork (Fig. B14 in Appendix B). There was no significant difference among different age groups in terms of locations or types of performed field activities (i.e., sampling practices, sample types, and the number of locations visited per field trip). However, equipment use differed significantly between age groups ($H_{(4, 172)} = 13.463$, $p = 0.009$; Fig. 2, Fig. B14 – B15 in Appendix B). Notably, significant differences were observed between students (age group 18–25), who used equipment most frequently in the highest number of different environments, and senior researchers (age group 56–65) who used it the least (Fig. 2). Furthermore, there was no significant correlation between the equipment use and the focus of fieldwork on invasive alien species.

3.3. Cleaning of field equipment, footwear, clothing, and transport

Clothes (80 %) and footwear (75 %) are often or always cleaned between field visits (Fig. B18 in Appendix B).

The most common method for cleaning field equipment, clothes, and footwear was the easiest one: visual inspection with manual removal of visible biomass (90–97 % of respondents), followed by drying and washing with detergents (clothes only; Figs B16, B19, B20 in Appendix B). Respondents usually use these two methods for cleaning field clothes and footwear after returning from field activities (50 %), while 32–36 % will apply the manual inspection between the field locations (i.e., before leaving the location; Figs B19, B20 in Appendix B). Most respondents (72–77 %) do not use disinfectants to clean their clothes or footwear (Figs B19, B20 in Appendix B). No significant differences in the

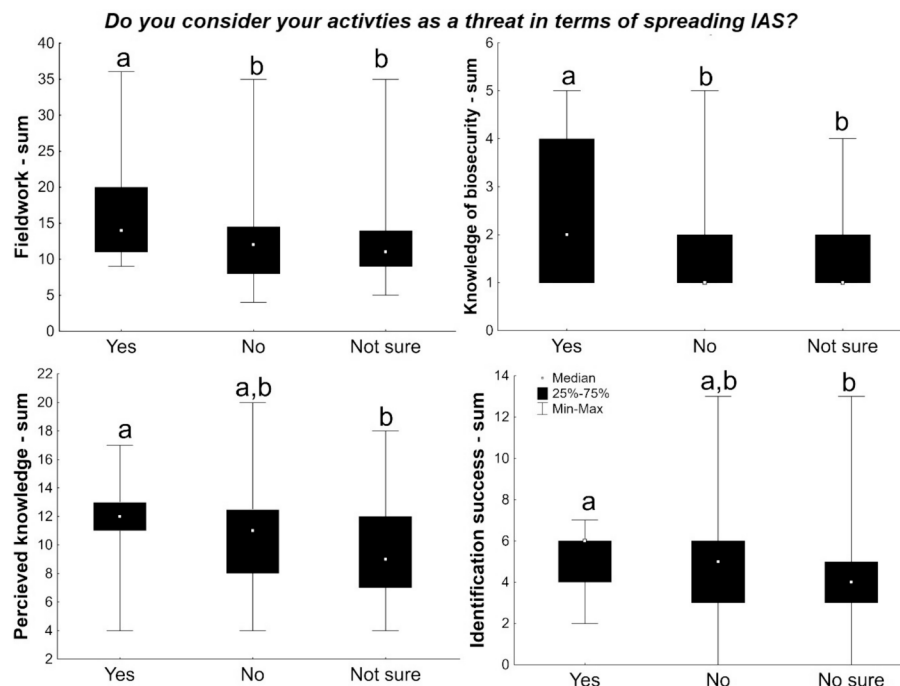


Fig. 1. Differences in respondents' perception of the risk of their fieldwork activities based on the frequency and scope performed fieldwork activities, knowledge of biosecurity, perceived knowledge of IAS, and identification success of IAS. Different letters depict significant differences.

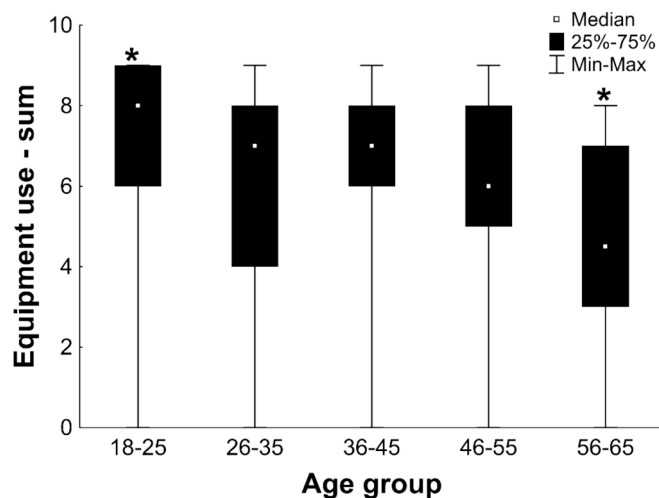


Fig. 2. Differences in equipment use (frequency and types of environments the equipment is used in) among different age groups. Groups that exhibit statistically significant differences are marked with an asterisk (*).

frequency and methods of clothing and footwear cleaning were observed between different age groups. However, clothing/footwear cleaning showed a positive correlation with the focus of fieldwork in IAS ($r_s = 0.151$, $p < 0.05$), indicating that respondents engaged in fieldwork related to IAS clean their clothing/footwear significantly more frequently and employ diverse methods.

Significant differences in the frequency and methods of equipment cleaning were identified between respondents from different age groups ($H_{(4, 172)} = 15.415$, $p = 0.004$; Fig. 3). Students (age group 18–25), who use the equipment most frequently, generally cleaned the equipment less often compared to other age groups, and significantly less compared to senior researchers (age group 46–55; Fig. 3). Furthermore, there was no significant correlation in the frequency and methods of equipment cleaning between respondents who are invasive species experts (i.e.,

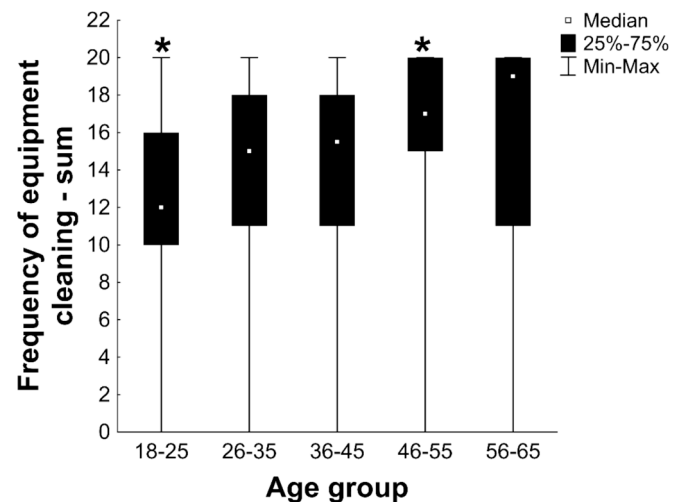


Fig. 3. Differences in the frequency of equipment cleaning among different age groups. Groups that display statistically significant differences are marked with an asterisk (*).

whose fieldwork is related to IAS) and those who are not.

Most respondents (38–60 %) never clean their vehicles when conducting fieldwork (Fig. B17 in Appendix B). Additionally, no significant differences were observed in the frequency and methods of vehicle cleaning between different age groups. Nevertheless, a positive correlation between vehicle cleaning and focus of fieldwork on IAS was identified ($r_s = 0.228$, $p < 0.05$), indicating that respondents whose fieldwork is related to IAS clean their vehicles significantly more frequently and employ diverse methods.

4. Discussion

Various human activities have been identified as a risk for the unintentional spread of IAS (Keller et al., 2011; Hulme, 2015; Saul et al.,

2017). However, some work-related activities, such as fieldwork conducted by researchers, remain largely understudied in this context. Biosecurity measures pertaining to fieldwork activities are seldom included in management or action plans for endangered and priority native species and habitats in Croatia, even though their regular monitoring is required by both national and EU regulations. While biosecurity measures are incorporated into existing management plans for IAS (which also require regular monitoring), they are only presented as recommendations. Monitoring of these species/habitats involves fieldwork, and researchers often find themselves needing to survey a higher number of locations with a limited amount of available equipment sets in a single monitoring event. Therefore, this research aimed to assess whether fieldwork could contribute to the accidental transportation and introduction of invasive alien species through equipment, clothing, footwear, or vehicles used during fieldwork. While only a limited number of studies has examined this issue thus far, the possibility of such transportation has been identified in most cases (Waterkeyn et al., 2010; Shannon et al., 2018, 2019, 2020a; Sutcliffe et al., 2018; Coughlan et al., 2020; Smith et al., 2020; LaFond et al., 2021). Consequently, it is also essential to evaluate field researchers' knowledge and perception of IAS, as well as their knowledge and application of biosecurity measures, in order to assess the actual threat they pose in terms of unintentional IAS spread.

According to the survey results, more than half of the respondents work in the fields of biology, ecology, or nature protection, implying they should possess at least a basic knowledge of biological invasions and IAS. This also poses a certain bias to the gathered data as the highly represented groups are expected to also have the highest awareness of IAS and biosecurity issues. This assumption was validated by the recorded success in identifying IAS: over 50 % of the respondents accurately identified five or more species out of seven displayed in the survey (Fig. B27 in Appendix B). This finding aligns with the research by Sutcliffe et al. (2018), which recorded a positive correlation between IAS knowledge and respondents' background, showing that participants with qualifications in ecology exhibit a higher level of IAS knowledge.

Regarding age and experience, our study identified the youngest age group (18–25 years), which roughly corresponds to the student experience group, to pose the highest risk for the unintentional spread of IAS. Students visited the highest number of various locations during fieldwork activities and used the equipment most frequently but at the same time displayed the lowest scores regarding its cleaning, indicating a general lack of awareness about this issue among this age group. This aligns with the results of the study by Shannon et al. (2019), in which students demonstrated a lower initial understanding of IAS biosecurity risks compared to professionals. This difference in awareness may reflect the lack of education on this topic during primary and secondary school (i.e., Ladrera et al., 2020; Sosa et al., 2021). A recent study by Sosa et al. (2021) established that the vast majority of professors believe biological invasions have less impact on biodiversity loss, both globally and locally, compared to other threats, such as habitat loss and degradation, pollution, and overexploitation of natural resources. Our results indicate that a equivalent gap in biosecurity education also exists at the university level. While students display similar levels of biosecurity knowledge, perceived knowledge of IAS, and identification success of IAS when compared to other age groups, their fieldwork practices were identified to pose a higher risk. Although Croatian universities offer undergraduate, master's, and PhD courses that cover biological invasions and include lectures on the ecological and socio-economic impacts of NNS and IAS, the general lack of formal biosecurity education is considered to significantly impact the perception of this issue (Piria et al., 2017; Piria et al., 2021). Moreover, considering the respondents' background in biology, we anticipate that they would be motivated to undertake biosecurity measures if they are aware of the issue. In addition to the lack of biosecurity education within the academic curriculum, the observed results may also stem from the overall lack of biosecurity regulations. The previously mentioned management plans

for several selected IAS in Croatia only include voluntary biosecurity measures and references to EU codes of conduct. Additionally, Croatia lacks nationwide biosecurity campaigns similar to initiatives such as the Check, Clean, Dry campaign, the Control of Aquatic Invasive Species and Restoration of Natural Communities in Ireland (CAISIE), or the Be Plant Wise campaign. These campaigns are designed to raise awareness and provide guidelines for reliable biosecurity practices among recreational water users and horticultural enthusiasts (Anderson et al., 2015; Rafuse, 2015; Hanrahan and Melly, 2019; Melly and Hanrahan, 2020; Smith et al., 2020). Apart from a recent media campaign that aimed to raise awareness about IAS in Croatia (MINGOR, 2022), both the general public and students are not exposed to biosecurity topics through media.

While most respondents cleaned their fieldwork clothes, footwear, and equipment, they did not employ various cleaning methods. Instead, the most commonly used method was the removal of visible biomass, which is the most basic also the least efficient cleaning approach. The financial costs associated with equipment sanitizing, along with the acquisition of additional equipment sets that would reduce the need for its frequent cleaning and reuse during fieldwork, could serve as significant barriers to improving biosecurity practices among both scientists and organizations (Sutcliffe et al., 2018). This could also explain why students are the least diligent in equipment cleaning.

The majority of respondents do not clean their vehicles before, during, or after conducting fieldwork activities (as depicted in Fig. B17 in Appendix B), despite vehicles serving as a significant pathway for the accidental transfer of IAS (Ascensão and Capinha, 2017). For instance, it has been established that the presence and abundance of roads connecting infested with uninfested areas increases the risk of spreading infection caused by the root pathogen *Phytophthora lateralis* (Jules et al., 2002; Floerl et al., 2009). Waterkeyn et al. (2010) highlighted the extent of vehicles' contribution to the unintentional IAS spread; their study demonstrated that car tires could transfer up to half a kilogram of mud containing an average of 33 propagules from various invertebrate species. Furthermore, a study by Hodkinson and Thompson (1997) revealed that 25 % of plant species discovered as seeds on cars were invasive to the research area. This fact clearly illustrates the importance of vehicle cleaning as a preventive measure against the spread of IAS and the urgent need for campaigns that raise awareness about the significance of this cleaning method.

Unfortunately, vehicle cleaning was also the most overlooked biosecurity measure in our study. Nevertheless, respondents whose fieldwork activities closely relate to IAS research demonstrated a significantly higher frequency of vehicle cleaning and employed a wider range of cleaning methods compared to others. This demonstrates that individuals who frequently work with IAS, and are therefore familiar with the topic, recognize the importance of meticulously cleaning all equipment used during fieldwork. Furthermore, these same respondents not only demonstrated the best results in terms of biosecurity practices, which refer to the practical implementation of biosecurity measures during the planning and execution of fieldwork, but also exhibited the most extensive knowledge about IAS. Their superior knowledge of biosecurity measures, better perceived knowledge of IAS, and more successful IAS identification, when compared to other respondents, is an important indicator that individuals who regularly engage with IAS are also aware of the significant risk associated with their potential spread. Consequently, they tend to adopt a more proactive stance in the implementation of robust biosecurity measures.

Similar results regarding the application of biosecurity measures and the perception of risks associated with the spread of IAS were observed in Shannon et al. (2019), though with some differences. According to their study, the respondents who exhibited a better risk perception still failed to adopt adequate biosecurity measures. Nevertheless, research related to IAS did exert a positive influence on risk perception and behaviour during fieldwork activities in the mentioned study. The already mentioned Check, Clean, Dry, and Be Plant Wise campaigns in the UK, along with general exposure to campaigns providing biosecurity

guidelines, have also positively impacted the risk perception of participants in their survey. Interestingly, the respondents who completed our questionnaire displayed a better perception of the risks they can pose regarding the dispersal of IAS compared to the participants in the Shannon et al. (2019) study, even in the absence of similar campaigns in Croatia. This result can potentially be attributed to a higher number of respondents in our survey (172 compared to 65 in Shannon et al., 2019), as well as to greater percentage of respondents working in the fields of biology, ecology, and related sciences in our study (57 % compared to 40 % in Shannon et al., 2019).

Regarding self-awareness, the respondents who perceived their activities a risk for IAS spread do indeed pose an actual and significant risk during their field activities: their equipment comes into contact with the widest variety of surfaces, they use the equipment on multiple locations per field trip while conducting several different activities per day, they collect the widest variety of samples that could potentially be transported between sites, and they visit the most sites per day. They also demonstrated the most comprehensive knowledge of biosecurity measures, which has the potential to effectively mitigate these risks. Furthermore, these participants consider their knowledge of IAS to be extremely good, a perception that is consistently mirrored in their success while identifying IAS. In contrast, the group that did not perceive their activities as a potential risk for IAS spread exhibited a significantly lower knowledge of biosecurity, their fieldwork activities were evaluated to be considerably less rigorous, while their perceived knowledge of IAS and their IAS identification success was highly variable.

Aside from knowledge, Sutcliffe et al. (2018) identified additional barriers to the implementation of biosecurity measures within one's work. These barriers include the absence of comprehensible guidelines, financial and time-related costs associated with performing biosecurity measures, as well as challenges in establishing collective responsibility and fostering attitude changes. Additionally, the lack of unified regulatory framework for biosecurity at both the EU and member state levels has also been highlighted as a significant barrier to biosecurity implementation by several studies (Baquero et al., 2021a; 2021b) which advocate the establishment of a centralized EU agency and a thoroughly integrated biosecurity system relying on cross-EU coordination and cooperation.

5. Conclusion

The results of our research demonstrate a generally insufficient knowledge and understanding of biosecurity practices. Only a small percentage of respondents, primarily those who work with IAS, applied more than just basic biosecurity measures connected to their fieldwork activities, and exhibited awareness of or participated in some form of biosecurity education. Biosecurity campaigns in Croatia are very much needed and they should i) firstly provide broader awareness-raising campaigns on this topic, and then ii) specific campaigns customized for particular stakeholder groups. Based on our results, in the case of researchers, such campaigns should especially emphasize vehicle cleaning procedures within biosecurity measures. They should also highlight the general importance of fieldwork planning to minimize potential biosecurity risks, the need to perform cleaning of all equipment, footwear and clothing frequently and by applying diverse methods. Also, a more comprehensive education regarding the biosecurity risks of IAS needs to be included in the academic curriculum, since students were identified as a high-risk group, potentially due to lack of awareness of the issue.

Furthermore, as mentioned throughout the literature, awareness raising campaigns are not a silver bullet solution to altering human behaviour (Wensing et al., 2010; Seimetz et al., 2016) and managing biosecurity risks posed by IAS. The so-called "knowing-doing gap" (Kitson and Straus, 2010), which indicates that significant behavioural changes require more than just improved knowledge and raised awareness, might lead to suboptimal performance of such interventions.

Hence, additional forms of interventions, particularly in governance and policy, will also be required. Some of these interventions are already present within the EU but only in the form of voluntary sectoral documents (i.e., EU codes of conduct for IAS; EASIN, 2023), or they are integrated as recommendations within specific national-level planning documents and lack comprehensive coordination on a larger scale. To enhance effectiveness, we share the perspective with Baquero et al. (2021a, 2021b) that obligatory governance and policy interventions should be incorporated as part of a more centralized and integrated biosecurity system, relying on cross-EU coordination and cooperation.

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CRedit authorship contribution statement

Lucija Novoselec: Data curation, Formal analysis, Investigation, Validation, Writing – original draft, Writing – review & editing, Methodology. **Lana Židak:** Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing, Visualization. **Arvena Bošnjak:** Data curation, Formal analysis, Validation, Writing – original draft. **Ana Depolo:** Data curation, Investigation, Methodology, Writing – original draft. **Mihaela Jakopčić:** Data curation, Investigation, Methodology, Validation, Writing – original draft. **Sandra Hudina:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jnc.2024.126648>.

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